

proj

2026-02-12

set up

```
library(tidyverse)
```

```
## —— Attaching core tidyverse packages ——
```

```
—— tidyverse 2.0.0 ——
```

```
## ✓ dplyr 1.2.0 ✓ readr 2.1.6
```

```
## ✓ forcats 1.0.1 ✓ stringr 1.5.2
```

```
## ✓ ggplot2 4.0.2 ✓ tibble 3.3.1
```

```
## ✓ lubridate 1.9.5 ✓ tidyr 1.3.2
```

```
## ✓ purrr 1.2.1
```

```
## —— Conflicts ——
```

```
—— tidyverse_conflicts() ——
```

```
## ✖ dplyr::filter() masks stats::filter()
```

```
## ✖ dplyr::lag() masks stats::lag()
```

```
## Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
```

```
library(skimr)
```

```
library(ggcorrplot)
```

```
library(broom)
```

```
library(sandwich)
```

```
library(lmtest)
```

```
## Loading required package: zoo
```

```
##
```

```
## Attaching package: 'zoo'
```

```
##
```

```
## The following objects are masked from 'package:base':
```

```
##
## as.Date, as.Date.numeric
```

```
set.seed(42)
```

importdata & checking

```
df_raw <- readr::read_csv("merged_data.csv", show_col_types = FALSE)
```

```
glimpse(df_raw)
```

```
## Rows: 12,228
```

```
## Columns: 27
```

```
## $ Date <date> 2019-07-05, 2019-07-06, 2019-07-0...
```

```
## $ Symbol <chr> "BTC", "BTC", "BTC", "BTC", "BTC",...
```

```
## $ Open <dbl> 11145.67, 10982.41, 11256.45, 1141...
```

```
## $ High <dbl> 11406.83, 11665.00, 11538.00, 1233...
```

```
## $ Low <dbl> 10796.44, 10964.51, 11094.37, 1122...
```

```
## $ Close <dbl> 10970.73, 11256.49, 11406.24, 1223...
```

```
## $ Volume <dbl> 63534.35, 51469.50, 38884.80, 5218...
```

```
## $ Price <dbl> 9.392662, 9.341968, 9.364348, 9.35...
```

```
## $ Log_Volume <dbl> 11.05934, 10.84874, 10.56836, 10.8...
```

```
## $ `Volatility of Intraday Price` <dbl> 0.04223248, 0.06145794, 0.03702299...
```

```
## $ `7D_EWMA_Intraday_Price_Volatility` <dbl> 0.07131592, 0.06885142, 0.060894
32...
```

```
## $ Return <dbl> -0.015820264, 0.025714034, 0.01321...
```

```
## $ Lagged_Return <dbl> -0.068843026, -0.015820264, 0.0257...
```

```
## $  $\sigma_{Rt}$  <dbl> 0.06715742, 0.05583598, 0.05420192...
```

```
## $ Amihud <dbl> 0.0014304895, 0.0023702313, 0.0012...
```

```
## $  $\sigma_{LIQt}$  <dbl> 0.003318641, 0.002910890, 0.002997...
```

```
## $ Log_Cap <dbl> 26.00792, 26.02211, 26.03225, 26.0...
```

```
## $ Datetime <date> 2019-07-05, 2019-07-06, 2019-07-0...
```

```
## $ BitcoinDominance <dbl> 64.68771, 64.72221, 64.66535, 65.2...
```

```
## $ RegSin <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
```

```
## $ RegEU <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
```

```
## $ Year_x      <dbl> 7.610358, 7.610358, 7.610358, 7.61...
## $ Month_x    <dbl> 1.94591, 1.94591, 1.94591, 1.94591...
## $ European_News_Index <dbl> 5.438079, 5.438079, 5.438079, 5.4
3...
## $ Year_y      <dbl> 7.610358, 7.610358, 7.610358, 7.61...
## $ Month_y     <dbl> 1.94591, 1.94591, 1.94591, 1.94591...
## $ SEPU        <dbl> 5.676954, 5.676954, 5.676954, 5.67...
```

```
skim(df_raw)
```

Data summary

Name	df_raw
Number of rows	12228
Number of columns	27

Column type frequency:

character	1
Date	2
numeric	24

Group variables	None
-----------------	------

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Symbol	0	1	3	4	0	6	0

Variable type: Date

skim_variable	n_missing	complete_rate	min	max	median	n_unique
Date	0	1	2019-07-05	2025-02-01	2022-04-19	2039
Datetime	0	1	2019-07-05	2025-02-01	2022-04-19	2039

Variable type: numeric

skim_variable	n_ mis sin g	comp lete_r ate	mea n	sd	p0	p2 5	p5 0	p75	p100	h i s t
Open	0	1	6184 .07	1.58 9957 e+04	0. 00	0.5 0	10 0.7 2	1816 .35	1.06 1438 e+05	
High	0	1	6331 .05	1.62 6547 e+04	0. 00	0.5 1	10 4.8 5	1848 .50	1.09 5880 e+05	
Low	0	1	6029 .36	1.55 2249 e+04	0. 00	0.4 8	96. 84	1777 .40	1.05 3215 e+05	
Close	0	1	6191 .68	1.59 2238 e+04	0. 00	0.5 0	10 0.7 2	1816 .89	1.06 1438 e+05	
Volume	0	1	3847 2780 7.19	2.16 6711 e+09	70 47 .9 0	29 46 17. 33	84 98 12. 25	2211 2877 9.00	1.09 0737 e+11	
Price	0	1	3.86	4.66 0000 e+00	- 6. 41	- 0.6 7	4.6 5	7.52	1.16 0000 e+01	
Log_Volume	0	1	15.1 7	3.65 0000 e+00	8. 86	12. 59	13. 65	19.2 1	2.54 2000 e+01	

skim_variable	n_ mis sin g	comp lete_r ate	mea n	sd	p0	p2 5	p5 0	p75	p100	h i s t
Log_Cap	0	1	24.2 4	1.99 0000 e+00	19 .0 9	22. 88	24. 12	25.9 1	2.83 8000 e+01	
BitcoinDominanc e	0	1	53.6 9	9.62 0000 e+00	39 .0 3	44. 14	53. 41	61.8 9	7.27 6000 e+01	
RegSin	0	1	0.00	2.00 0000 e-02	0. 00	0.0 0	0.0 0	0.00	1.00 0000 e+00	
RegEU	0	1	0.00	2.00 0000 e-02	0. 00	0.0 0	0.0 0	0.00	1.00 0000 e+00	
Year_x	0	1	7.61	0.00 0000 e+00	7. 61	7.6 1	7.6 1	7.61	7.61 0000 e+00	
Month_x	0	1	1.69	7.40 0000 e-01	0. 00	1.3 9	1.9 5	2.30	2.48 0000 e+00	
European_News _Index	0	1	5.62	2.60 0000 e-01	4. 95	5.4 4	5.6 2	5.79	6.18 0000 e+00	

skim_variable	n_ mis sin g	comp lete_r ate	mea n	sd	p0	p2 5	p5 0	p75	p100	h i s t
Year_y	0	1	7.61	0.00 0000 e+00	7. 61	7.6 1	7.6 1	7.61	7.61 0000 e+00	
Month_y	0	1	1.69	7.40 0000 e-01	0. 00	1.3 9	1.9 5	2.30	2.48 0000 e+00	
SEPU	0	1	5.59	2.10 0000 e-01	5. 20	5.4 6	5.5 7	5.72	6.15 0000 e+00	

```
na_summary <- df_raw %>%
  summarise(across(everything(), ~sum(is.na(.)))) %>%
  pivot_longer(everything(), names_to = "col", values_to = "na_count") %>%
  arrange(desc(na_count))
```

```
na_summary
```

```
## # A tibble: 27 × 2
```

```
##   col          na_count
##   <chr>         <int>
## 1 Date           0
## 2 Symbol         0
## 3 Open           0
## 4 High           0
## 5 Low            0
## 6 Close          0
## 7 Volume         0
## 8 Price          0
```

```
## 9 Log_Volume          0
## 10 Volatility of Intraday Price    0
## #      17 more rows
```

BTC

```
library(tidyverse)
```

```
library(lubridate)
```

```
library(skimr)
```

```
df <- df_raw %>%
  filter(Symbol == "BTC") %>%
  mutate(
    date = as.Date(Date),
    year = year(date),
    month = month(date),
    wday = wday(date, label = TRUE, week_start = 1),
    is_weekend = wday %in% c("Sat", "Sun")
  ) %>%
  arrange(date)
```

```
glimpse(df)
```

```
## Rows: 2,039
## Columns: 32
## $ Date          <date> 2019-07-05, 2019-07-06, 2019-07-0...
## $ Symbol        <chr> "BTC", "BTC", "BTC", "BTC", "BTC",...
## $ Open          <dbl> 11145.67, 10982.41, 11256.45, 1141...
## $ High          <dbl> 11406.83, 11665.00, 11538.00, 1233...
## $ Low           <dbl> 10796.44, 10964.51, 11094.37, 1122...
## $ Close         <dbl> 10970.73, 11256.49, 11406.24, 1223...
## $ Volume        <dbl> 63534.35, 51469.50, 38884.80, 5218...
## $ Price         <dbl> 9.392662, 9.341968, 9.364348, 9.35...
## $ Log_Volume    <dbl> 11.05934, 10.84874, 10.56836, 10.8...
```

```
## $ `Volatility of Intraday Price`      <dbl> 0.04223248, 0.06145794, 0.03702299...
## $ `7D_EWMA_Intraday_Price_Volatility` <dbl> 0.07131592, 0.06885142, 0.060894
32...
## $ Return                             <dbl> -0.015820264, 0.025714034, 0.01321...
## $ Lagged_Return                       <dbl> -0.068843026, -0.015820264, 0.0257...
## $  $\sigma_{Rt}$                      <dbl> 0.06715742, 0.05583598, 0.05420192...
## $ Amihud                             <dbl> 0.0014304895, 0.0023702313, 0.0012...
## $  $\sigma_{LIQt}$                    <dbl> 0.003318641, 0.002910890, 0.002997...
## $ Log_Cap                             <dbl> 26.00792, 26.02211, 26.03225, 26.0...
## $ Datetime                           <date> 2019-07-05, 2019-07-06, 2019-07-0...
## $ BitcoinDominance                   <dbl> 64.68771, 64.72221, 64.66535, 65.2...
## $ RegSin                             <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
## $ RegEU                              <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
## $ Year_x                             <dbl> 7.610358, 7.610358, 7.610358, 7.61...
## $ Month_x                            <dbl> 1.94591, 1.94591, 1.94591, 1.94591...
## $ European_News_Index                 <dbl> 5.438079, 5.438079, 5.438079, 5.4
3...
## $ Year_y                             <dbl> 7.610358, 7.610358, 7.610358, 7.61...
## $ Month_y                            <dbl> 1.94591, 1.94591, 1.94591, 1.94591...
## $ SEPU                               <dbl> 5.676954, 5.676954, 5.676954, 5.67...
## $ date                               <date> 2019-07-05, 2019-07-06, 2019-07-0...
## $ year                               <dbl> 2019, 2019, 2019, 2019, 2019, 2019...
## $ month                              <dbl> 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7...
## $ wday                               <ord> Fri, Sat, Sun, Mon, Tue, Wed, Thu,...
## $ is_weekend                         <lgl> FALSE, TRUE, TRUE, FALSE, FALSE, F...
```

```
skim(df %>% select(date, year, month, wday, is_weekend))
```

Data summary

Name	df %>% select(date, year,...
Number of rows	2039
Number of columns	5

Column type frequency:

Date	1
factor	1
logical	1
numeric	2

Group variables None

Variable type: Date

skim_variable	n_missing	complete_rate	min	max	median	n_unique
date	0	1	2019-07-05	2025-02-01	2022-04-19	2039

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
wday	0	1	TRUE	7	Fri: 292, Sat: 292, Mon: 291, Tue: 291

Variable type: logical

skim_variable	n_missing	complete_rate	mean	count
is_weekend	0	1	0.29	FAL: 1456, TRU: 583

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
year	0	1	2021.78	1.64	2019	2020	2020	2020	2020	
month	0	1	6.70	3.49	1	4	7	10	12	

seems good

```
names(df)
```

```
## [1] "Date"                "Symbol"
## [3] "Open"                "High"
## [5] "Low"                 "Close"
## [7] "Volume"              "Price"
## [9] "Log_Volume"          "Volatility of Intraday Price"
## [11] "7D_EWMA_Intraday_Price_Volatility" "Return"
## [13] "Lagged_Return"       " $\sigma_{Rt}$ "
## [15] "Amihud"               " $\sigma_{LIQt}$ "
## [17] "Log_Cap"              "Datetime"
## [19] "BitcoinDominance"     "RegSin"
## [21] "RegEU"                "Year_x"
## [23] "Month_x"               "European_News_Index"
## [25] "Year_y"                "Month_y"
## [27] "SEPU"                  "date"
## [29] "year"                  "month"
## [31] "wday"                  "is_weekend"
```

```
df <- df %>%
```

```
  mutate(vol = ` $\sigma_{Rt}$ `)
```

```
summary(df$vol)
```

```
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.002338 0.018061 0.025827 0.029414 0.036461 0.229135
```

```
ggplot(df, aes(x = vol)) +
```

```
  geom_histogram(bins = 60) +
```

```
  labs(
```

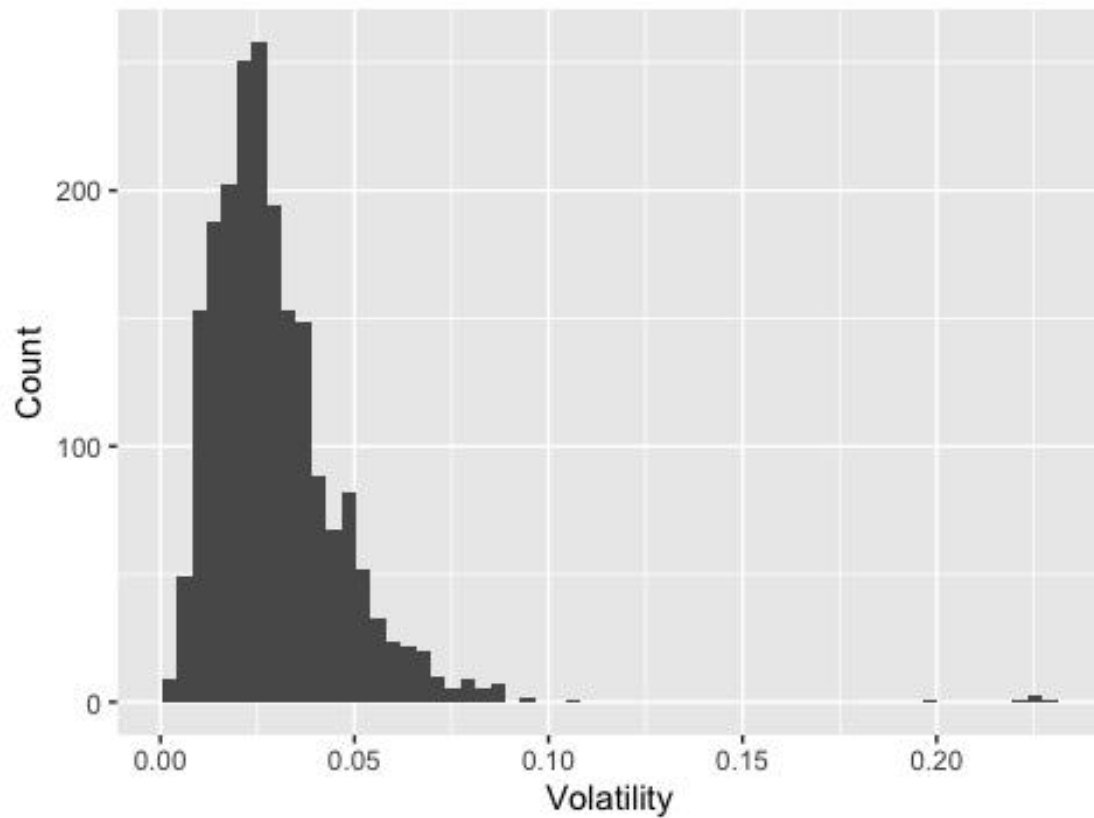
```
    title = "3.1 Distribution of Bitcoin Volatility",
```

```
    x = "Volatility",
```

```
    y = "Count"
```

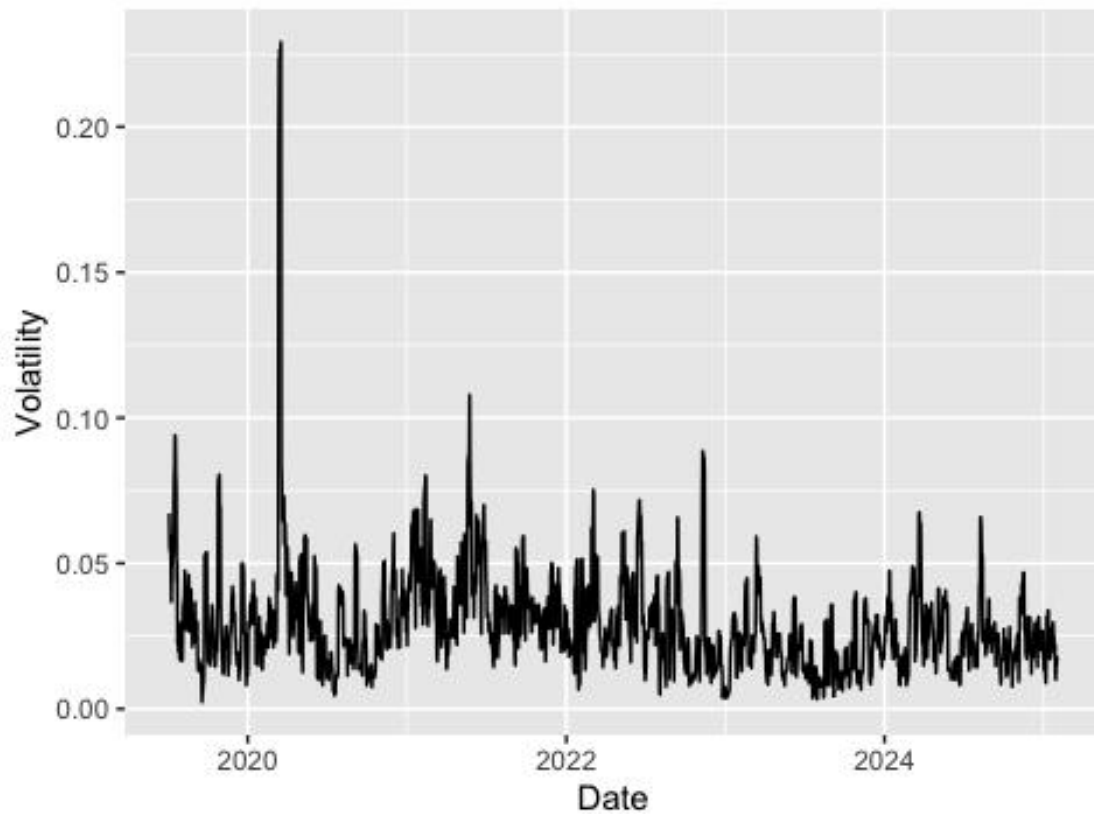
```
)
```

3.1 Distribution of Bitcoin Volatility



```
ggplot(df, aes(x = date, y = vol)) +  
  geom_line() +  
  labs(  
    title = "3.2 Bitcoin Volatility Over Time",  
    x = "Date",  
    y = "Volatility"  
  )
```

3.2 Bitcoin Volatility Over Time



corr analysis

```
df_analysis <- df %>%
```

```
  select(
    date,
    vol,
    log_volume = Log_Volume,
    volume = Volume,
    news = European_News_Index,
    sepu = SEPU,
    btc_dom = BitcoinDominance,
    ret = Return
  )
```

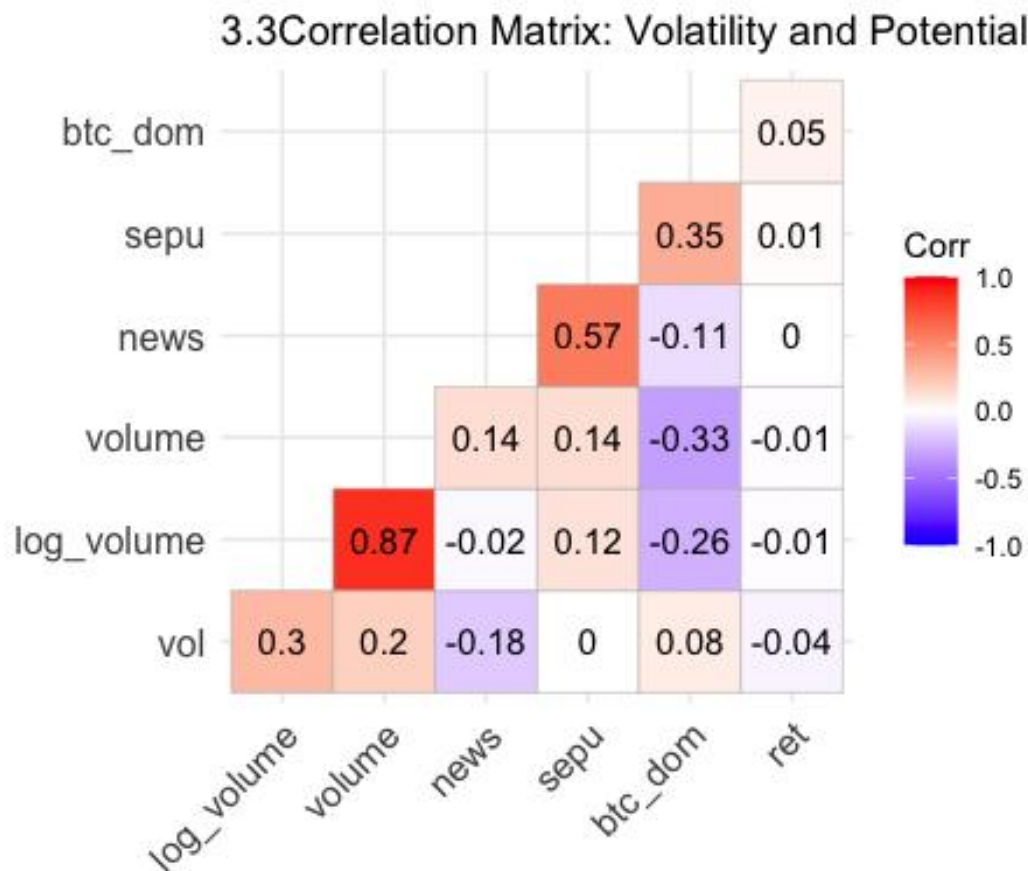
```
library(ggcorrplot)
```

```
corr_mat <- cor(
```

```
df_analysis %>% select(-date),
use = "pairwise.complete.obs"
)
```

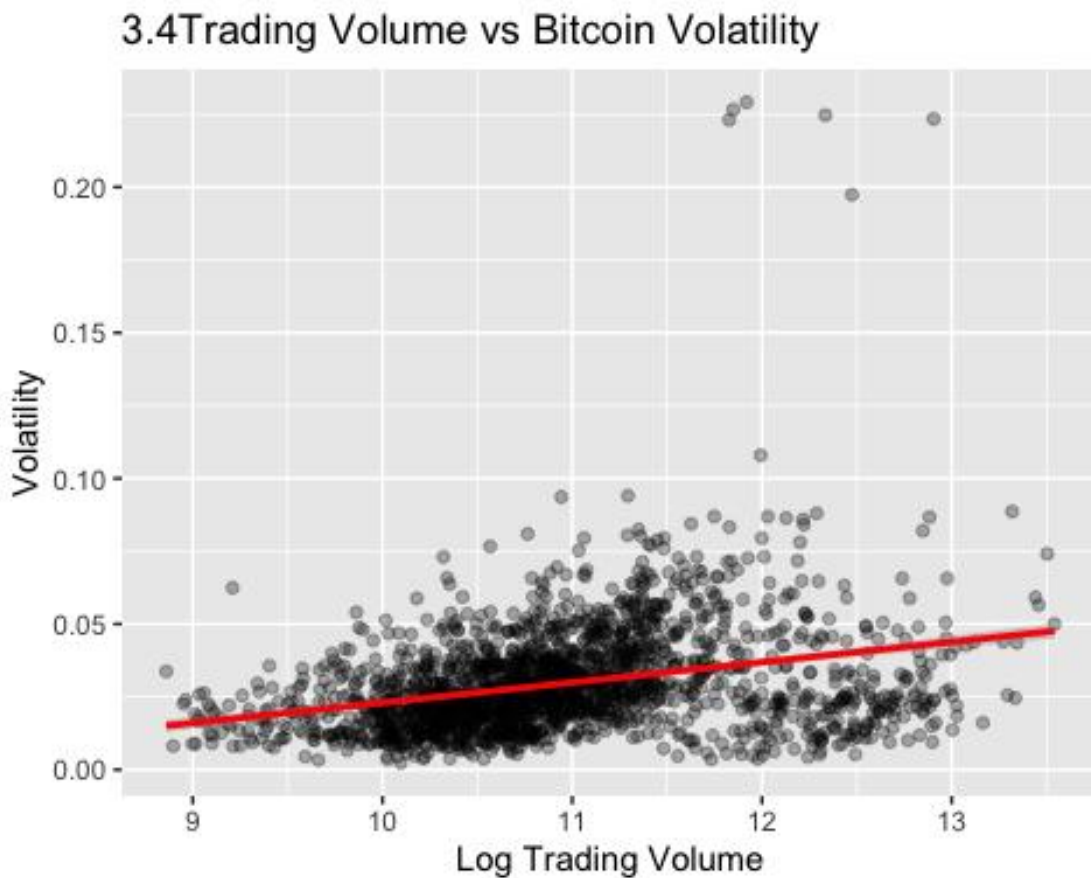
```
ggcorrplot(corr_mat, type = "lower", lab = TRUE) +
  ggtitle("3.3Correlation Matrix: Volatility and Potential Factors")
```

```
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## Please use tidy evaluation idioms with `aes()`.
## See also `vignette("ggplot2-in-packages")` for more information.
## The deprecated feature was likely used in the ggcorrplot package.
## Please report the issue at <https://github.com/kassambara/ggcorrplot/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



```
ggplot(df_analysis, aes(x = log_volume, y = vol)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", color = "red") +
  labs(
    title = "3.4Trading Volume vs Bitcoin Volatility",
    x = "Log Trading Volume",
    y = "Volatility"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```
corr_rank <- df_analysis %>%
  select(-date) %>%
  summarise(across(everything(), ~cor(.x, df_analysis$vol, use = "pairwise.complete.
obs")))) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "corr_with_vol")
```

```
%>%
  filter(variable != "vol") %>%
  mutate(abs_corr = abs(corr_with_vol)) %>%
  arrange(desc(abs_corr))
```

corr_rank

```
## # A tibble: 6 × 3
##   variable  corr_with_vol abs_corr
##   <chr>      <dbl>    <dbl>
## 1 log_volume    0.301    0.301
## 2 volume       0.200    0.200
## 3 news        -0.176    0.176
## 4 btc_dom      0.0797   0.0797
## 5 ret         -0.0396   0.0396
## 6 sepu        0.00125  0.00125
```

```
knitr::kable(
  corr_rank,
  caption = "Table 3.1: Correlation Ranking of Potential Influencing Factors"
)
```

Table 3.1: Correlation Ranking of Potential Influencing Factors

variable	corr_with_vol	abs_corr
log_volume	0.3005876	0.3005876
volume	0.1996265	0.1996265
news	-0.1763940	0.1763940
btc_dom	0.0797242	0.0797242
ret	-0.0395952	0.0395952
sepu	0.0012460	0.0012460

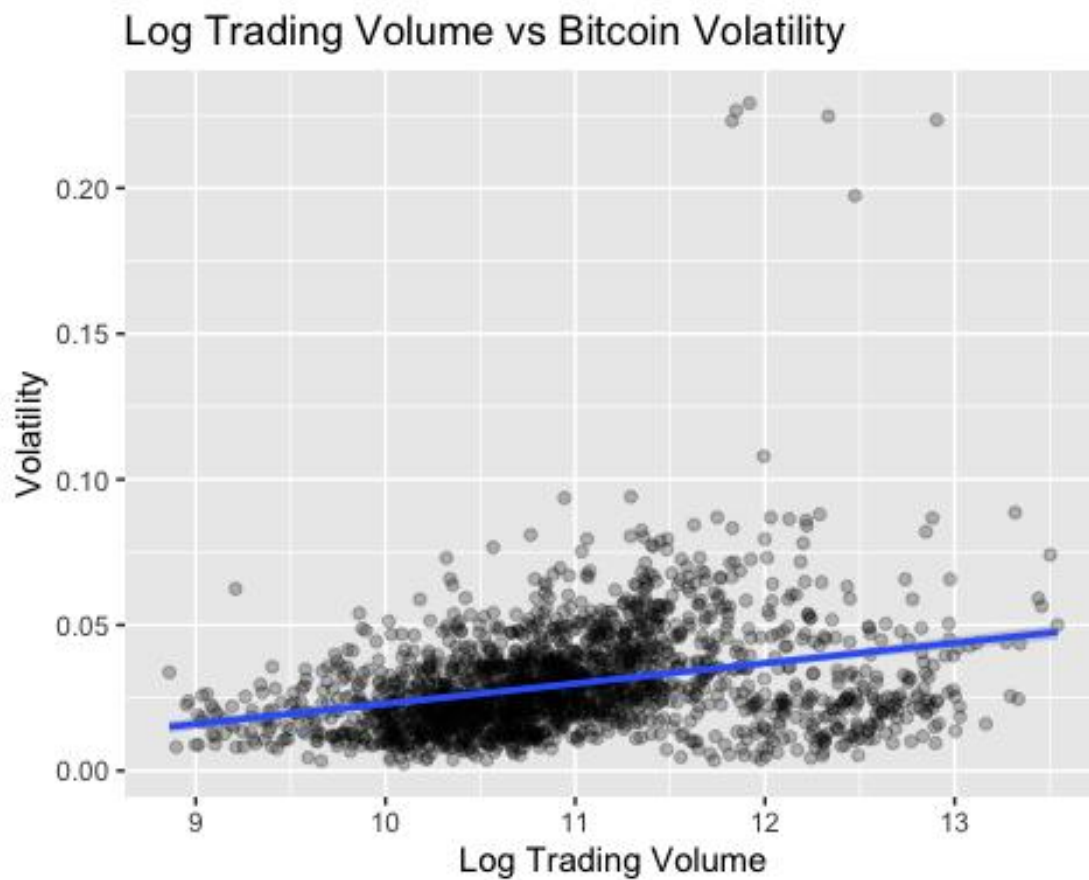
```
ggplot(df_analysis, aes(x = log_volume, y = vol)) +
  geom_point(alpha = 0.25) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
```

```

title = "Log Trading Volume vs Bitcoin Volatility",
x = "Log Trading Volume",
y = "Volatility"
)

## `geom_smooth()` using formula = 'y ~ x'

```

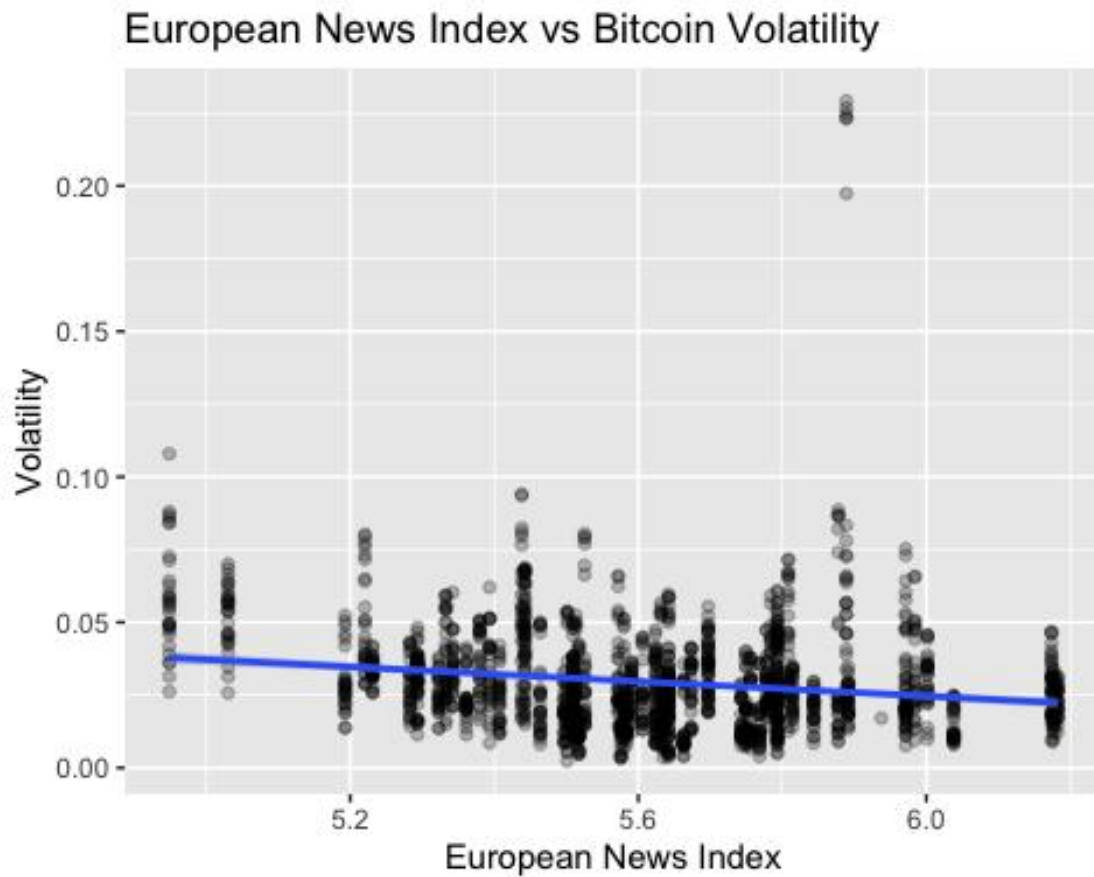


```

ggplot(df_analysis, aes(x = news, y = vol)) +
  geom_point(alpha = 0.25) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "European News Index vs Bitcoin Volatility",
    x = "European News Index",
    y = "Volatility"
  )

```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```
library(dplyr)
```

```
library(lmtest)
```

```
library(sandwich)
```

```
library(stargazer)
```

```
##
```

```
## Please cite as:
```

```
## Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.
```

```
## R package version 5.2.3. https://CRAN.R-project.org/package=stargazer
```

```
df_model <- df_analysis %>%
```

```
  mutate(
```

```

log_volume_l1 = lag(log_volume, 1),
news_l1 = lag(news, 1),
sepu_l1 = lag(sepu, 1),
btc_dom_l1 = lag(btc_dom, 1),
ret_l1 = lag(ret, 1)
) %>%
drop_na()

m1 <- lm(vol ~ log_volume_l1 + news_l1 + sepu_l1 + btc_dom_l1 + ret_l1, data = df_model)

summary(m1)

##
## Call:
## lm(formula = vol ~ log_volume_l1 + news_l1 + sepu_l1 + btc_dom_l1 +
##     ret_l1, data = df_model)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.031502 -0.010713 -0.001963  0.007398  0.188538
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1.145e-02  1.137e-02  -1.006   0.314
## log_volume_l1  8.232e-03  5.244e-04  15.698 < 2e-16 ***
## news_l1       -9.718e-03  2.019e-03  -4.814 1.59e-06 ***
## sepu_l1       -2.118e-03  2.701e-03  -0.784   0.433
## btc_dom_l1     3.243e-04  5.039e-05   6.436 1.52e-10 ***
## ret_l1        -4.524e-02  1.095e-02  -4.132 3.75e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
##
## Residual standard error: 0.01707 on 2032 degrees of freedom
## Multiple R-squared:  0.1547, Adjusted R-squared:  0.1526
## F-statistic: 74.38 on 5 and 2032 DF,  p-value: < 2.2e-16

coeftest(m1, vcov. = NeweyWest(m1, lag = 5, prewhite = FALSE, adjust = TRUE))

##
## t test of coefficients:
##
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.01144649  0.03262143 -0.3509  0.72571
## log_volume_l1  0.00823218  0.00163656  5.0302 5.331e-07 ***
## news_l1      -0.00971848  0.00499973 -1.9438  0.05206 .
## sepu_l1      -0.00211828  0.00475724 -0.4453  0.65617
## btc_dom_l1    0.00032434  0.00015294  2.1207  0.03407 *
## ret_l1       -0.04523624  0.03034206 -1.4909  0.13615
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

stargazer(m1, type="text")

##
## =====
##           Dependent variable:
##           -----
##                   vol
## -----
## log_volume_l1      0.008***
##                   (0.001)
##
## news_l1            -0.010***
##                   (0.002)
##
```

```

## sepu_l1          -0.002
##                  (0.003)
##
## btc_dom_l1       0.0003***
##                  (0.0001)
##
## ret_l1           -0.045***
##                  (0.011)
##
## Constant         -0.011
##                  (0.011)
##
## -----
## Observations      2,038
## R2                0.155
## Adjusted R2       0.153
## Residual Std. Error 0.017 (df = 2032)
## F Statistic       74.380*** (df = 5; 2032)
## =====
## Note:             *p<0.1; **p<0.05; ***p<0.01

```